

Introduction to Digital Signal Processing

HW6: Optimal FIR and Short-time Fourier Transform

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June 14, 2026

Part I: DFT and FFT

Exercise 1 (DFT and FFT Properties)

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(a) $x[n] = \sin\left(\frac{n\pi}{8}\right), 0 \leq n \leq 7 \Rightarrow x[n] = [0, 1, 0, -1, 0, 1, 0, -1]$

$X[k] = \sum_{n=0}^7 x[n] e^{-jk\frac{2\pi}{8}n} = e^{-jk\frac{\pi}{4}} + e^{-jk\frac{3\pi}{4}} + e^{-jk\frac{5\pi}{4}} + e^{-jk\frac{7\pi}{4}} \Rightarrow X[k] = [0, 0, -4j, 0, 0, 0, 4j, 0]$

$x[n] = \frac{1}{8} \sum_{k=0}^7 X[k] e^{jk\left(\frac{2\pi}{8}\right)n}$ $X[k] = \begin{cases} -4j & k=2 \\ 4j & k=6 \\ 0 & \text{o.w.} \end{cases}$

$\parallel$

$\frac{1}{2j} \left(e^{j \cdot 2 \cdot \left(\frac{2\pi}{8}\right) \cdot n} - e^{j \cdot (-2) \cdot \left(\frac{2\pi}{8}\right) \cdot n} \right) \stackrel{\uparrow}{=} \frac{1}{8} \left(-4j e^{j \cdot 2 \cdot \left(\frac{2\pi}{8}\right) \cdot n} + 4j e^{j \cdot 6 \cdot \left(\frac{2\pi}{8}\right) \cdot n} \right)$

(b) for $N=8$ point FFT (2 Cooley-Tukey algorithm) complex multiplications $= \frac{N}{2} \log_2 N = 4 \times 3 = 12$

(c) The order is determined by bit-reversal permutation

$x[0]$	0000	$\longrightarrow$	0000	$X[0]$
$x[1]$	0001	$\longrightarrow$	1000	$X[2]$
$x[2]$	0010	$\longrightarrow$	0100	$X[4]$
$x[3]$	0011	$\longrightarrow$	1100	$X[6]$
$x[4]$	0100	$\longrightarrow$	0010	$X[1]$
$x[5]$	0101	$\longrightarrow$	1010	$X[3]$
$x[6]$	0110	$\longrightarrow$	0110	$X[5]$
$x[7]$	0111	$\longrightarrow$	1110	$X[7]$
$x[8]$	1000	$\longrightarrow$	0001	$X[9]$
$x[9]$	1001	$\longrightarrow$	1001	$X[11]$
$x[10]$	1010	$\longrightarrow$	0101	$X[13]$
$x[11]$	1011	$\longrightarrow$	1101	$X[15]$
$x[12]$	1100	$\longrightarrow$	0011	$X[10]$
$x[13]$	1101	$\longrightarrow$	1011	$X[12]$
$x[14]$	1110	$\longrightarrow$	0111	$X[14]$
$x[15]$	1111	$\longrightarrow$	1111	$X[15]$

Part II: Information Retrieval and Optimal FIR Design

Exercise 2 (*Exercise 2: Dual-tone Multiple Frequency (DTMF)*)

(a) Window Length and Frequency Resolution Analysis

In this experiment, we adjusted the window length L for a Hann window to observe its effect on frequency resolution. With a sampling frequency $f_s = 8000$ Hz, we compared the spectral responses for three window lengths: $L = 160$ (0.02s), $L = 80$ (0.01s), and $L = 40$ (0.005s).

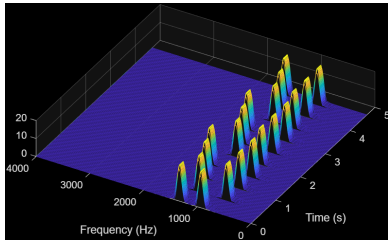


Figure 1: $L = 160$ (0.02s)

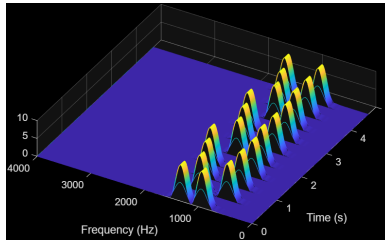


Figure 2: $L = 80$ (0.01s)

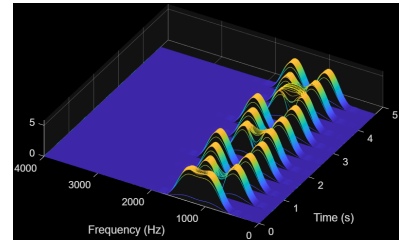


Figure 3: $L = 40$ (0.005s)

Figure 4: Comparison of 3D spectrograms of DTMF signals with different Hann window lengths.

Observations and Discussion:

- **Frequency Resolution vs. Window Length:** Decreasing L causes spectrogram peaks to broaden significantly. According to $\Delta f \approx 1/T$, shorter windows limit frequency resolution, causing main-lobe broadening and increased spectral leakage.
- **Resolution Threshold:** While $L = 160$ and $L = 80$ clearly distinguish dual tones with a visible energy valley, $L = 40$ (0.005s) leads to peak merging, creating a flat-topped response where tones become unresolvable.
- **Failure Point:** Digit '7' (852 Hz and 1209 Hz) was the first to fail at $L = 40$, indicating that extremely short windows compromise the system's ability to separate specific frequency pairs.

Comparison of Window Functions: We further investigated the resolution limits of different window functions (Rectangular, Hann, and Blackman) by determining the minimum window length L required to resolve the DTMF dual tones.

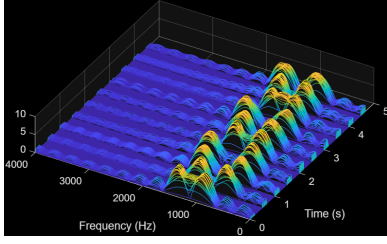


Figure 5: Rectangular ($L = 25$)

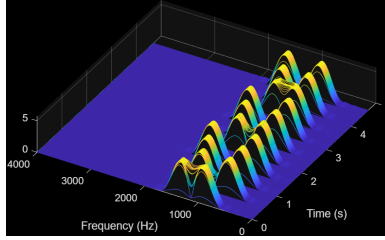


Figure 6: Hann ($L = 44$)

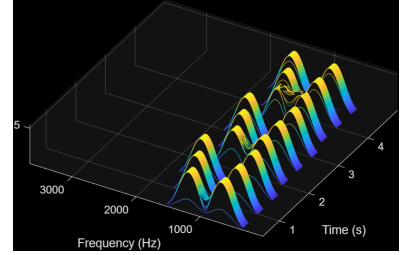


Figure 7: Blackman ($L = 52$)

Figure 8: Minimum window lengths for frequency resolution: Rectangular vs. Hann vs. Blackman.

- **Resolution Performance:** The Rectangular window achieves the highest frequency resolution (minimum $L = 25$) due to its narrowest main-lobe width ($\approx 4\pi/L$). However, it exhibits significant spectral leakage due to high side-lobes.
- **Trade-off Analysis:** The Blackman window provides superior side-lobe suppression for cleaner spectra but requires the longest window ($L = 52$) because of its wide main-lobe ($\approx 12\pi/L$).
- **Conclusion:** The Hann window ($L = 44$) acts as a balanced design, offering an effective compromise between resolution and leakage suppression, making it highly suitable for DTMF signal analysis.

(b) Signal Decoding and Identification

To decode the DTMF sequence, we performed spectral analysis on the received signal. We identified the signal's dominant frequencies (F_1, F_2) at eight specific time intervals to determine the dialed digits based on the standard DTMF frequency matrix.

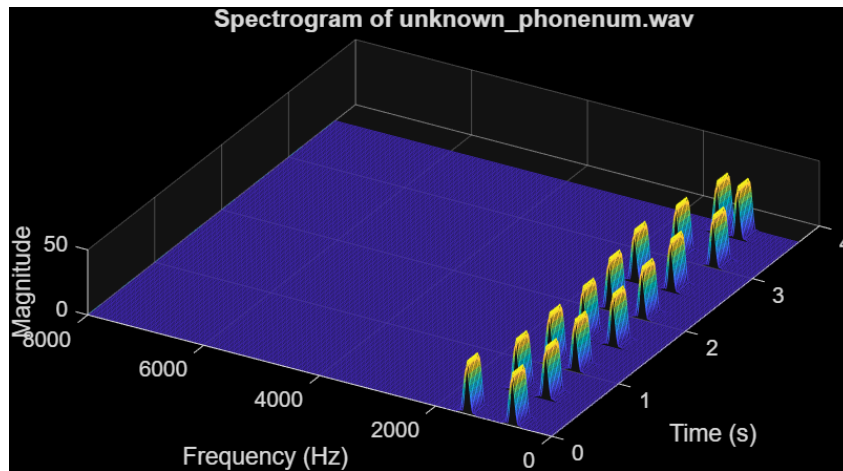


Figure 9: Overall spectrogram of the unknown DTMF signal sequence.

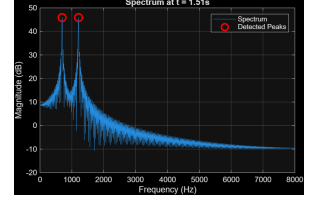
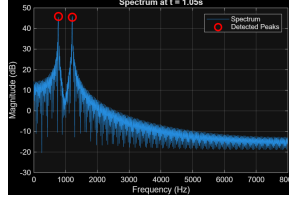
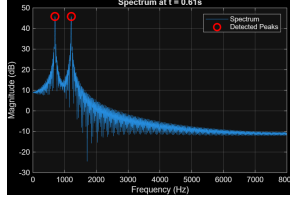
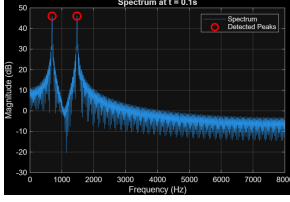


Figure 10: $t = 0.10$ s

Figure 11: $t = 0.61$ s

Figure 12: $t = 1.05$ s

Figure 13: $t = 1.51$ s

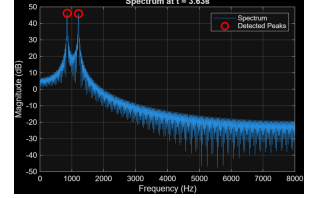
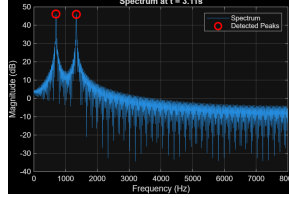
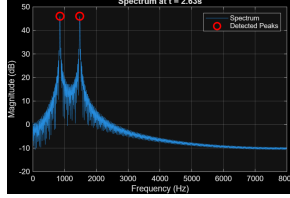
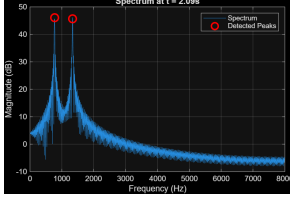


Figure 14: $t = 2.09$ s

Figure 15: $t = 2.63$ s

Figure 16: $t = 3.11$ s

Figure 17: $t = 3.63$ s

Figure 18: Individual spectral analysis at identified time intervals.

- **Detection Methodology:** We can employed three approaches: (1) Manual spectral inspection via FFT peaks, (2) Automatic frequency detection using the `findpeaks` algorithm, and (3) Auditory identification.
- **Digit Identification Table:** The following table summarizes the identified frequency pairs and their corresponding DTMF digits by employing method (1).

Table 1: Summary of DTMF decoding results.

Time (s)	F_1 (Hz)	F_2 (Hz)	Detected Digit
0.10	695	1477	3
0.61	695	1211	1
1.05	773	1211	4
1.51	695	1211	1
2.09	773	1336	5
2.63	852	1477	9
3.11	695	1336	2
3.63	852	1211	7

- **Automatic Detection:** We implemented a robust MATLAB decoding algorithm using Short-Time Fourier Transform (STFT). By utilizing the `findpeaks` function, we extracted dominant frequencies and matched them against the standard DTMF matrix. To ensure reliability, we incorporated a de-bouncing mechanism that filters transient noise by requiring a detected digit to persist across multiple consecutive time frames:

Listing 1: MATLAB de-bouncing logic.

```
% Robust automatic decoding with de-bouncing logic
detected_digits = []; last_digit = -1; consecutive_count = 0;
for col = 1:length(t)
    [pks, locs] = findpeaks(abs(S(:, col)), 'SortStr', 'descend', 'NPeaks', 2);
    if length(locs) < 2, continue; end
    f_peaks = sort(freqs(locs));
    [~, r] = min(abs(low_freqs - f_peaks(1)));
    [~, c] = min(abs(high_freqs - f_peaks(2)));
    if r <= 4 && c <= 3 && keypad(r, c) ~= -1
        current_digit = keypad(r, c);
        if current_digit == last_digit
            consecutive_count = consecutive_count + 1;
        else
            if consecutive_count > 3, detected_digits = [
                detected_digits, last_digit]; end
            last_digit = current_digit; consecutive_count = 1;
        end
    end
end
if consecutive_count > 3, detected_digits = [detected_digits,
    last_digit]; end
```

- **Accuracy:** All three methods yielded consistent results, confirming the sequence **{3, 1, 4, 1, 5, 9, 2, 7}**. The automatic detection method demonstrated high reliability, effectively filtering noise and identifying the exact peak frequencies within the specified time windows.

Exercise 3 (*Optimal FIR Design via Parks-McClellan*)

The Parks-McClellan algorithm is implemented to iteratively approximate an optimal low-pass filter with equal ripple characteristics. The parameters for all experiments are defined as follows:

- **Original Parameters:** $\omega_c = 0.35\pi$, $\omega_s = 0.4\pi$, $x_c = \cos(\omega_c)$, $x_s = \cos(\omega_s)$, and $\text{tol} = 1e-6$.
- **Initial Alternating Points:** The initial guess for the alternating points is:

```
omega_init = [0.05 0.12 0.17 omega_c/pi omega_s/pi 0.48 0.7
    0.92 1]*pi;
```

(a) Initial Guess and Convergence

- **Convergence Criterion:** The algorithm employs a convergence tolerance of $\text{tol} = 1e-6$. The while loop terminates when the maximum absolute change between consecutive iterations of alternating points falls below this threshold:

$$\max |x_{\text{current}} - x_{\text{prev}}| < \text{tol} \quad (1)$$

- **Experimental Setup:** We tested three different initial configurations of alternating points to evaluate the convergence speed of the Remez exchange algorithm. The initial guesses are defined as:

```
% 1. Standard (9 points)
omega_init = [0.05 0.12 0.17 omega_c/pi omega_s/pi 0.48 0.7
              0.92 1]*pi;

% 2. Dense (12 points)
omega_init = [0.03 0.05 0.12 0.17 0.26 omega_c/pi omega_s/pi
              0.5 0.65 0.7 0.92 1]*pi;

% 3. Ultra-Dense (16 points)
omega_init = [0.03 0.05 0.12 0.14 0.17 0.26 0.30 omega_c/pi
              omega_s/pi 0.5 0.65 0.7 0.75 0.81 0.92 1]*pi;
```

- **Observations and Results:** The convergence performance under the specified $\text{tol} = 1\text{e-}6$ is summarized below:

Configuration	Number of Points	Iterations	Final δ
Standard	9	6	0.2213
Dense	12	8	0.1656
Ultra-Dense	16	9	0.1094

- **Discussion:** The results confirm the trade-off between approximation precision and computational complexity. By enforcing a strict $\text{tol} = 1\text{e-}6$, the algorithm ensures that the Remez exchange process reaches a highly stable state. As the number of points increases, the algorithm gains more degrees of freedom to minimize the Minimax error (δ), resulting in a superior equiripple response. However, the increased dimensionality of the **BigMatrix** search space necessitates more iterations to align the extrema, as reflected in our data.

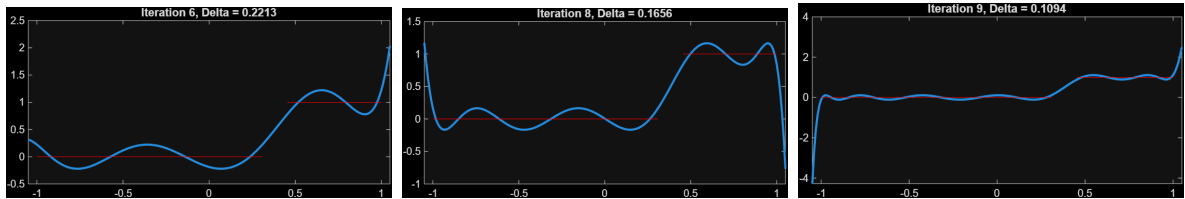


Figure 19: Evolution of equiripple approximation: 9-point (left), 12-point (center), and 16-point (right) configurations.

- **Conclusion:** Higher point density effectively reduces the maximal approximation error (δ), enabling a flatter frequency response. While the computational cost (iterations) rises, the strict tolerance of 10^{-6} ensures that the final design converges to the globally optimal Minimax solution, confirming the effectiveness of the Parks-McClellan approach.

(b) Impact of Cut-off Frequency (ω_c)

- **Experimental Setup:** To investigate the sensitivity of the Parks-McClellan algorithm to the cut-off frequency, we performed a parameter sweep of ω_c from 0.35π to 0.44π , while keeping the stop-band edge ω_s fixed at 0.45π . The algorithm was executed with a fixed tolerance of `tol = 1e-6`.
- **Observations and Results:** The following figures illustrate the relationship between the cut-off frequency and both the final approximation error (δ) and the required iterations for convergence.

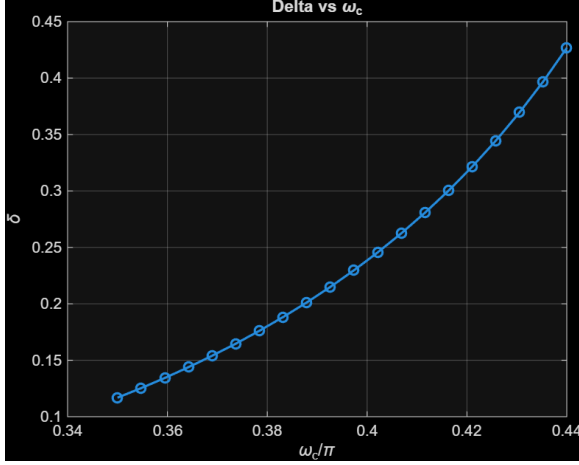


Figure 20: Maximal approximation error (δ) vs ω_c .

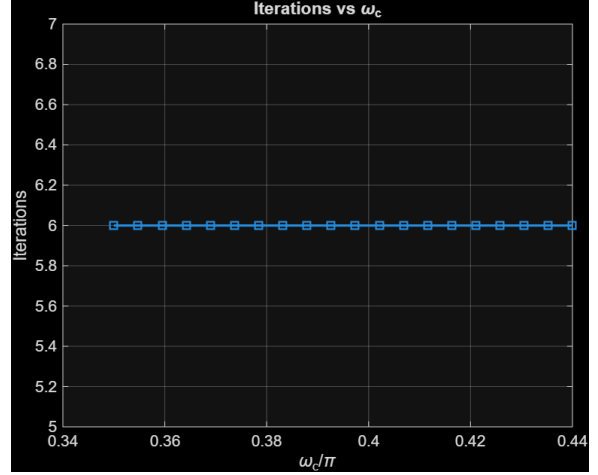


Figure 21: Number of iterations vs ω_c .

- **Discussion:**
 - **Approximation Error (δ vs ω_c):** As ω_c increases, the pass-band of the low-pass filter expands, reducing the transition bandwidth ($\omega_s - \omega_c$). Our results show that δ increases monotonically with ω_c . This occurs because, with a fixed filter order, expanding the pass-band requires the filter to approximate a wider frequency range, which inherently leads to larger ripples to maintain the equiripple property in the Minimax sense.
 - **Convergence Stability (Iterations vs ω_c):** The iteration count remains constant at 6 across the entire range of tested ω_c . This indicates that the Parks-McClellan algorithm exhibits robust convergence characteristics for this specific filter order, as the structural complexity of the Remez exchange remains consistent despite the shifting frequency boundaries.
- **Conclusion:** Varying the cut-off frequency significantly impacts the approximation error (δ) due to the trade-off between the pass-band width and the Minimax ripple magnitude. However, the convergence speed of the algorithm remains remarkably stable, demonstrating its effectiveness in maintaining a consistent design process even as the frequency specifications evolve.

(c) Explanation of the Increasing Delta Value

- **Phenomenon:** It is observed that the maximal approximation error (δ) increases during the initial iterations. This behavior is a fundamental characteristic of the Remez exchange process, reflecting the algorithm's path toward the global Minimax solution.

- **Theoretical Basis:** The progression of δ is governed by the **Chebyshev Alternation Theorem**:

1. **Initial Guess Suboptimality:** The algorithm begins with an initial set of points $\{x_i\}$ that do not coincide with the true extrema of the error function $E(x)$, leading to an initial underestimation of the Minimax error.
2. **Proof by Contradiction (Existence of Larger Error):** Assume the current calculated error δ is already the global minimum (the Minimax error), but the point set $\{x_i\}$ is not optimal. This implies there exists an alternative polynomial $P^*(x)$ with error $\delta^* < \delta$. Define the difference polynomial $D(x) = P^*(x) - P(x)$.
 - At the $L + 2$ alternating points $\{x_i\}$, $P(x_i) = H_d(x_i) - \sigma_i \delta$.
 - Since $\delta^* < \delta$, the sign of $P^*(x_i) - P(x_i)$ must be the same as the sign of the error σ_i .
 - By the sign-alternating property, $D(x)$ must have at least $L + 1$ zeros within the interval.
 - However, $D(x)$ is a polynomial of degree at most L . A degree- L polynomial with $L + 1$ zeros must be identically zero ($D(x) \equiv 0$), which contradicts the assumption that $P(x) \neq P^*(x)$.

Thus, if the current point set is not optimal, there must exist points where the error magnitude exceeds the current δ .

3. **Refinement via Remez Exchange:** The algorithm identifies these points where the error exceeds the current δ , performs an "exchange" to update the extrema, and iteratively "levels" the ripple magnitudes toward δ_{opt} .
- **Conclusion:** The increase in δ signifies that the algorithm is iteratively identifying the true extrema of the error function. By mathematically requiring the error to rise until all $L + 2$ extrema are equal in magnitude, the process ensures convergence to the globally optimal equiripple filter design.